Projective Informational Framework (MPFM)

A Structural Approach to Physical Reality

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Abstract

We introduce a projective informational framework in which information is considered primitive, while geometry, time, and physical dynamics emerge from a non-faithful and irreversible projection process.

In this framework, any observation, interaction, or measurement is modeled as a projection that irreversibly destroys part of the pre-projective informational content. The accumulation of information loss across successive projections leads to a natural hierarchy of scales, the emergence of an arrow of time, effective geometric structures, and robust forms characterized by an effective spectral dimension lower than that of their apparent support.

The proposed framework does not introduce new particles, new forces, or ad hoc degrees of freedom. Instead, it provides a structural interpretation of existing physical theories—quantum mechanics, general relativity, turbulence, and cosmology—viewed as effective descriptions valid within specific projection regimes, while clarifying their domains of validity and structural limitations.

This work is presented as a foundational and axiomatic–conceptual framework rather than as a phenomenological predictive model. It is open to critical examination, comparison with other informational, geometric, and entropic approaches, and indirect confrontation with existing and future experimental observations.

Keywords

Fundamental information; Non-faithful projection; Emergence of time; Spectral dimension; Fractal structure; Informational gravity; Quantum mechanics; General relativity; Turbulence; Cosmology; Foundations of physics

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